

Koysul Amin

Lead Software Test Engineer

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SUMMARY

Quality Assurance Engineer with 15+ years of expertise in automated and manual testing across web, client-server, and Windows applications. Deep proficiency in Selenium, PyTest, Python, and K6. Daily practitioner of Spec-Driven Development — writing detailed test specifications first, then using Claude Code to implement robust automation from them. Focused on shift-left testing, CI/CD integration, and accelerating quality through AI-assisted workflows.

EXPERIENCE

Lead Software Test Engineer · Dealertrack Technologies

Aug 2013 – Present · New York, NY

- Built a comprehensive end-to-end automation framework in Python and PyTest for functional, regression, and data-transformation testing across web and client-server applications.
- Validated complex data transformations — JSON→JSON and JSON→XML — for high-volume integration payloads.
- Implemented a shift-left testing approach: functional tests run inside the main codebase through the CI/CD pipeline, from pull-request creation through production deployment.
- Developed simulation APIs and service virtualization to mimic external dependencies, enabling isolated, reliable test environments.
- Conducted REST API testing and performance testing with K6, monitoring behavior under load to identify bottlenecks.
- Provisioned on-demand test environments on AWS to scale automation infrastructure.
- Leverage Claude Code (CLI) daily for spec-driven test generation, intelligent debugging, and rapid automation development; use Cursor IDE for AI-assisted review and refactoring.
- Collaborate in Agile/Scrum teams across multiple releases (Rally), ensuring comprehensive coverage and timely delivery.

Software Engineer · Prometric

Jul 2012 – Feb 2013 · Baltimore, MD

- Authored and executed manual and automated test cases in Microsoft Test Manager, validating business requirements against expected results.
- Designed data-driven regression suites with QuickTest Professional (QTP) for efficient, repeatable coverage.
- Drove defect triage and analysis with development teams to prioritize and resolve high-severity issues.

QA Analyst · AstraZeneca

Aug 2011 – Jul 2012 · Waltham, MA

- Performed end-to-end testing — functional, GUI, backend, and regression — for global pharmacology applications.
- Maintained requirement-to-test traceability matrices in HP Quality Center, ensuring coverage of critical functionality.
- Partnered with developers on defect analysis, significantly reducing production defects and improving application stability.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, JSON, XML, HTML

API & Performance: REST, Postman, SoapUI, K6, LoadRunner

Practices: Shift-left testing, CI/CD, service virtualization, data validation, Agile/Scrum

Databases: Oracle, SQL Server, DynamoDB, Redis

Test Automation: PyTest, Selenium WebDriver, framework design, API mocking

AI-Assisted QA: Claude Code (CLI), Spec-Driven Development, Cursor IDE, prompt engineering

Platforms: AWS, Oracle WebLogic, Apache Tomcat, Linux/Unix, Windows, macOS

Test Management: HP Quality Center, Rally, Jira, TFS

EDUCATION & CERTIFICATIONS

B.S., Technical Management · DeVry University, New York, NY GPA 3.67

A.S., Telecommunications Engineering · CUNY — New York City College of Technology, Brooklyn, NY

ISTQB Certified Tester, Foundation Level (CTFL) · Intl. Software Testing Qualifications Board